

ClimMaTech - IRMEWS

Intelligent River Monitoring & Early Warning System



Technical Capability & System Specification Sheet

System Overview

The **ClimMaTech Intelligent River Monitoring System (IRMEWS)** is a field-deployable, solar-powered, AI + IoT solution for **real-time hydrological monitoring, flood prediction, and early warning** in riverine and reservoir environments.

Designed and validated at **IIT Mandi**, the system integrates radar-based level and velocity sensors, rainfall and temperature monitoring, imaging modules, and cloud analytics — enabling **continuous, non-contact, and intelligent river data acquisition**.

Hardware & Sensor Specifications

Component	Description	Key Features	Range / Accuracy
Radar Level Sensor (CLM-LR05)	61 GHz FMCW radar, non-contact	Measures river water level	Range 0–50 m (customizable) Accuracy ± 1 mm Resolution 0.5 mm
Radar Velocity Sensor (CLM-LVR05)	24 GHz K-band Doppler radar	Measures surface flow velocity	Range 0.02–20 m/s Accuracy $\pm 1\%$ of measured value Resolution 0.01 m/s
Rainfall & Temperature Module	Tipping bucket (0.2 mm tip) + thermistor	Measures site rainfall & ambient temp	Rainfall up to 500 mm/h Temp $-20\text{ }^{\circ}\text{C}$ to $+70\text{ }^{\circ}\text{C}$
Camera (CLM-CAM01)	RPi wide angle camera module	Captures surface visuals & turbidity	1080p @ 1 fps interval IR support for low light
Edge Processor (CLM-EDGE02)	Raspberry Pi 5 with Relay Module and other sensors	Performs ML inference and system control	Supports auto ON/OFF to save power 5 V DC input

Communication Module	GSM / LoRa / NB-IoT / Wi-Fi	Transmits data and alerts to cloud	Encrypted data push via MQTT or HTTPS
Power Subsystem	Solar Panel (40–80 W) + Battery (12 V LiFePO4)	Off-grid operation with charge controller & buck converter	Runtime > 3 days autonomy Low power < 1.5 W active
Enclosure	IP68 polycarbonate or aluminum housing	Field rugged for high humidity & debris conditions	Operating Temp: –20 °C to +70 °C

Software & Analytics Capabilities

Feature	Description
Edge-AI Filtering	On-device ML filtering of radar signals for noise reduction and adaptive calibration.
Predictive ML Model (CLM-ML01)	Forecasts 1–3 hour water-level trends using time-series + rainfall correlation.
Digital Twin Platform (VisiFlow™)	3D visualization of flood impact using UAV data, LiDAR, and camera feed.
Dashboard & Alerts (Aagah™)	Real-time web dashboard for NHPC & SDMA control rooms; tiered alert levels with SMS/IVR integration.
Data Storage	MongoDB / Timescale DB on secure server for archiving and analytics.
Integration API	MODBUS, MQTT, RS-485, and REST API for SCADA or third-party systems.
Security	End-to-end encryption with TLS 1.3 and device IDs.

Communication & Integration

- Multi-protocol IoT connectivity: RS-485 / MODBUS / MQTT / LoRa / NB-IoT.
- Data accessible via:
 - **ClimMaTech Cloud Dashboard** (web & mobile)
 - **Customer control room interface**
 - **Integration with various portals** on request.
- Edge–cloud hybrid design ensures local operation during network outages.

Pilot Validation (UHL River, Kamand, Mandi – 2025)

Metric	Performance
System Uptime	98.5 %
Alert Lead Time	15 - 20 minutes depends on upstream devices location
Depth Accuracy	± 1 mm (verified vs manual gauge)
Velocity Accuracy	± 1%
Data Latency	< 60 seconds (edge-to-cloud)
Operational Continuity	> 3 days autonomous without sunlight - can be increased with additional batteries

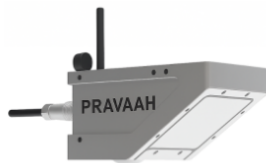
Advantages

- **Non-Contact Measurement:** Radar-based sensing unaffected by water quality, debris, or temperature fluctuations.
- **High Accuracy & Resolution:** Millimeter-level precision in level and velocity measurement, even under turbulent flow.
- **Edge AI Processing:** Embedded analytics for real-time filtering and adaptive signal calibration.
- **Multi-Protocol Connectivity:** Supports RS-485, MODBUS, LoRa, and MQTT for seamless IoT and SCADA integration.
- **Rugged & Reliable:** IP68-rated housing built for harsh environmental and high-humidity conditions.
- **Solar & Battery Compatible:** Low power draw enables easy deployment in remote and off-grid locations.
- **Scalable Integration:** Fully compatible with **ClimMaTech's Digital Twin Platform**, cloud dashboard, and Aagah™ early warning system.
- **Easy Installation & Maintenance:** Lightweight, top-mounted sensors with minimal calibration requirements.



Technical Specification of AWLR System

Specification	Requirement	ClimMaTech Device Specification
Make	—	ClimMaTech Pvt. Ltd.
Model No.	—	CLM-LR05 (Radar-Based Level Sensor)
Measuring Principle	Ultrasonic / Non-contact	61 GHz FMCW Radar (W-Band) Non-contact
Range	0 – 5 m	0 – 50 m (customizable)
Accuracy	1 mm	± 1 mm
Resolution	—	0.5 mm
Beam Angle	—	8° Azimuth × 8° Elevation
Sampling Frequency	—	1 sps (default) / up to 10 sps configurable
Output Signal	4 – 20 mA	4 – 20 mA / RS-485 / RS-232 / SDI-12 / LoRa / MQTT
Ingress Protection	IP68	IP68 (fully sealed polycarbonate enclosure)
Operating Voltage	12 V	9 – 30 V DC (solar compatible)
Power Consumption	—	< 1.5 W active / 0.2 W standby
Material	PVDF	PVDF sensor housing
Temperature Range	–20 °C to +70 °C	–25 °C to +70 °C (without heating)
Communication Interface	—	LoRa / NB-IoT / Wi-Fi / GSM (optional)
Mounting Type	—	Top-mount / bridge mount with adjustable bracket
Data Integration	—	Compatible with ClimMaTech Cloud and Digital Twin Platform



Technical Specification of AWL&VR System

Specification	Requirement	ClimMaTech Device Specification
Make	—	ClimMaTech Pvt. Ltd.
Model No.	—	CLM-LVR05 (Radar Surface Velocity Sensor)
Measuring Principle	Non-contact Doppler Radar	24 GHz K-Band Doppler Radar (FMCW)
Detection Distance	—	Up to 40 m above water surface
Velocity Range	0.05 – 15 m/s	0.02 – 20 m/s
Resolution	—	0.01 m/s
Accuracy	± 2 %	± 1 % of measured value
Beam Angle	—	12° Azimuth × 30° Elevation
Output Signal	4 – 20 mA / RS-485	4 – 20 mA, RS-485, RS-232, LoRa, MQTT
Digital Outputs	—	Open Collector (50 V / 200 mA)
Ingress Protection	IP68	IP68 (Waterproof & dust-proof housing)
Operating Voltage	12 V	9 – 30 V DC (solar / battery compatible)
Material	PVDF / Polycarbonate	Aluminum body with PVDF antenna shield
Temp. Range	–20 °C to +70 °C	–30 °C to +70 °C
Communication Protocols	—	ASCII-S, MODBUS-RTU, GLX-NMEA, SDI-12
Mounting Type	—	Bridge, pole, or wall mount (adjustable tilt)
Data Integration	—	Direct sync with ClimMaTech Dashboard & Digital Twin Engine
Special Features	—	Edge-AI filtering for noise reduction, auto self-calibration, solar ready



Technical Specification of AWL&VR + Camera Sensor

Specification	ClimMaTech Device Specification
Make	ClimMaTech Pvt. Ltd.
Model No.	CLM-CVDR05 (Vision + Radar Module)
Function	Dual-mode optical monitoring for flow velocity, surface profiling, and rainfall tracking
Camera Type	Full HD (1080p) AI-enabled industrial camera module
Lens & Field of View	2.8–12 mm varifocal lens / ~120° horizontal FOV
Frame Rate	30 fps (standard), 60 fps burst mode
Measurement Range	Ideal for river/drain width up to 20 m
Velocity Estimation	0.02 m/s to 14 m/s via optical flow AI tracking and Pravaah sensor cross-validation
Rainfall Monitoring	Integrated tipping-bucket rain gauge
Verification Mode	Dual validation – optical AI velocity & radar velocity
Output Signal	RS-485 / MODBUS / Ethernet / Wi-Fi / LoRa / MQTT
Data Storage	On-board microSD (up to 128 GB) + cloud sync
Power Supply	12 V DC / Solar panel (60 W recommended)
Power Consumption	< 4 W active / 0.5 W standby
Ingress Protection	IP67 (industrial outdoor enclosure)
Material	Aluminum body + polycarbonate lens cover
Operating Temperature	–20 °C to +70 °C
Mounting	Bridge, pole, or channel-side mount (adjustable tilt)
Communication Protocols	MODBUS-RTU, HTTP API, NTP-time sync for multi-node networking
Integration	Seamless connectivity with ClimMaTech Cloud & Aagah™ Early Warning Platform



Technical Specification of Early Warning System

Specification	ClimMaTech Device Specification
Make / Model	ClimMaTech Pvt. Ltd. – (CLM-EWS-01)
Function	Community-level flood / disaster early warning siren system
Activation Source	Remote wireless trigger from devices / Aagah™ EWS platform / manual trigger
Alert Types	Multi-tone siren + programmable voice message playback
Sound Output Level	120–130 dB @ 1 m (effective reach up to 500 m radius)
Coverage Range	Up to 50 km network coverage (via LoRa / LTE)
Communication	LoRa / MQTT / GSM / 4G LTE (configurable redundancy)
Operating Modes	Automatic (sensor-triggered) / Semi-auto (command-based)
Response Time	< 10 seconds from signal receipt to siren activation
Power Source	12 V DC solar system (with solar charge controller)
Solar Panel Rating	20-50 W (customizable per site)
Battery Backup	12 V, 20 Ah LiFePO4 – up to 7 days autonomy
Power Consumption	Idle: < 1 W / Active alert: < 30 W
Enclosure Rating	IP66 / IP67 weatherproof housing
Operating Temp.	–20 °C to +70 °C
Mounting Options	Pole / rooftop mount with vibration damping clamps
Connectivity	SIM-based 4G / LoRa node / RS-485
Local Indicators	LED beacon flash, status buzzer, and system health LED
Integration	Seamlessly connects to ClimMaTech Cloud , Aagah™ , and Digital Twin systems

Technical Specification of Data Logger

Specification	ClimMaTech Device Specification
Make / Model	ClimMaTech Pvt. Ltd. — CLM-DL01
Core Compute	Raspberry Pi 5 (quad-core ARM Cortex-A76, 64-bit OS) + Arduino (ATmega328/ESP32 option) for real-time IO and watchdog
Custom Electronics	Custom PCB for power routing, sensor conditioning, galvanic isolation, relay control, and peripheral interfaces
Storage	microSD 128 GB (onboard) + optional USB SSD expansion
Sensor Interfaces	SDI-12 (master), RS-485 (Modbus RTU), analog inputs (0–5 V / 4–20 mA via shunt), pulse/counter, UART, I2C, SPI
Communications	4G LTE module (primary) + NB-IoT option, LoRaWAN / private LoRa, Wi-Fi, GigE Ethernet
Telemetry Protocols	MQTT over TLS, HTTPS/REST, SFTP, Modbus-RTU/TCP, NTP time sync
Power Supply	DC input 9–30 V; integrated solar charge controller
Power Management	Low-power sleep modes; handles deep sleep; controlled power to peripherals for extreme low-power profiles
Relays & Outputs	Programmable relays (NO/NC) for siren/siren activation (Rakshak) or remote control
Real-time IO	Sensors handles high-priority interrupts, pulse counting, and failsafe I/Os
Edge Processing	On-device Python/C pipelines; ability to run TensorFlow Lite models on Pi5 for event detection & filtering
Security	Secure boot options, device authentication (certs), TLS1.2/1.3, encrypted storage (optional)
Environmental	IP67/IP68 enclosure option; operating –20 °C to +70 °C
Timekeeping	RTC (battery backed) + GPS time sync (optional)
Mounting	Pole / bridge / wall mount brackets; vibration damping
Integration	Native connector to ClimMaTech Cloud, Digital Twin, Aagah™ APIs; configurable data retention & sync policies

Key Innovations

- **Non-Contact Radar Measurement:** Immune to debris, sediment, or temperature changes.
- **Edge-AI Self-Calibration:** Maintains precision without frequent manual intervention.
- **Digital Twin Visualization:** 3D simulation of flood and flow dynamics. (optional)
- **Sustainable Power:** Solar + battery operation for remote, off-grid deployments.
- **Local Resilience:** Designed & manufactured in Himachal Pradesh; optimized for Himalayan terrain.

Deployment & Service Model

- **Mounting:** Bridge / pier / riverbank installation.
 - **Power:** Solar panel with charge controller and battery backup.
 - **Data Access:** Via ClimMaTech dashboard login or API feed.
 - **Maintenance:** Quarterly field inspection; remote diagnostics via cloud.
 - **Support:** Local technical service team (IIT Mandi region).
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- **Make in India / Atmanirbhar-compliant design**

Contact & Support



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